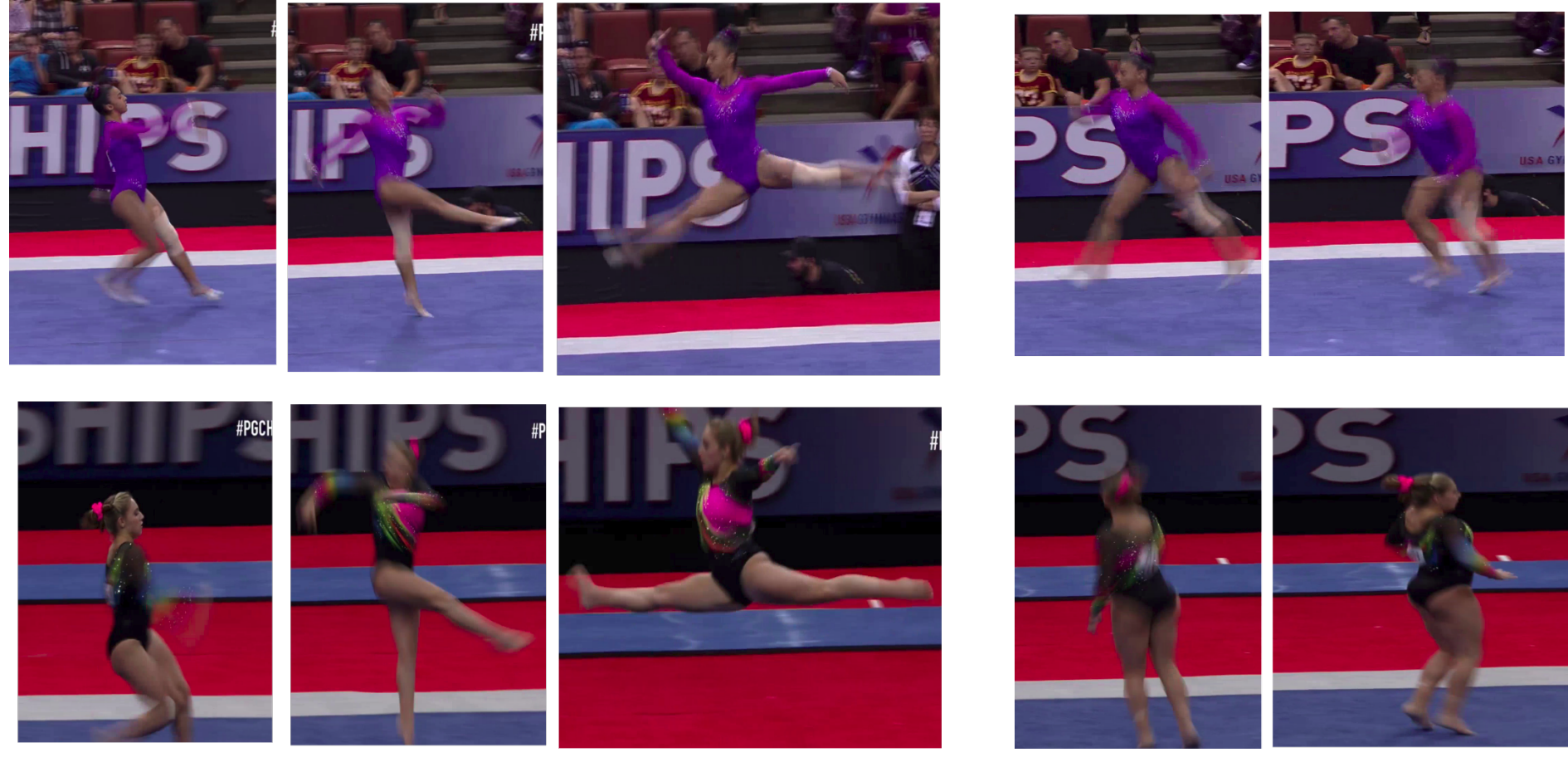


1. Motivation



Switch leap: 0.5-turn vs. 1-turn

Same posture, same trajectory — separable only by rotation.

- Fine-grained cues are **subtle**; appearance is near-identical.
- Multimodal SotA needs **pose / text / flow** → annotation & compute cost.

Research Question

Can **RGB-only** match multimodal SotA?

3. Key Contribution

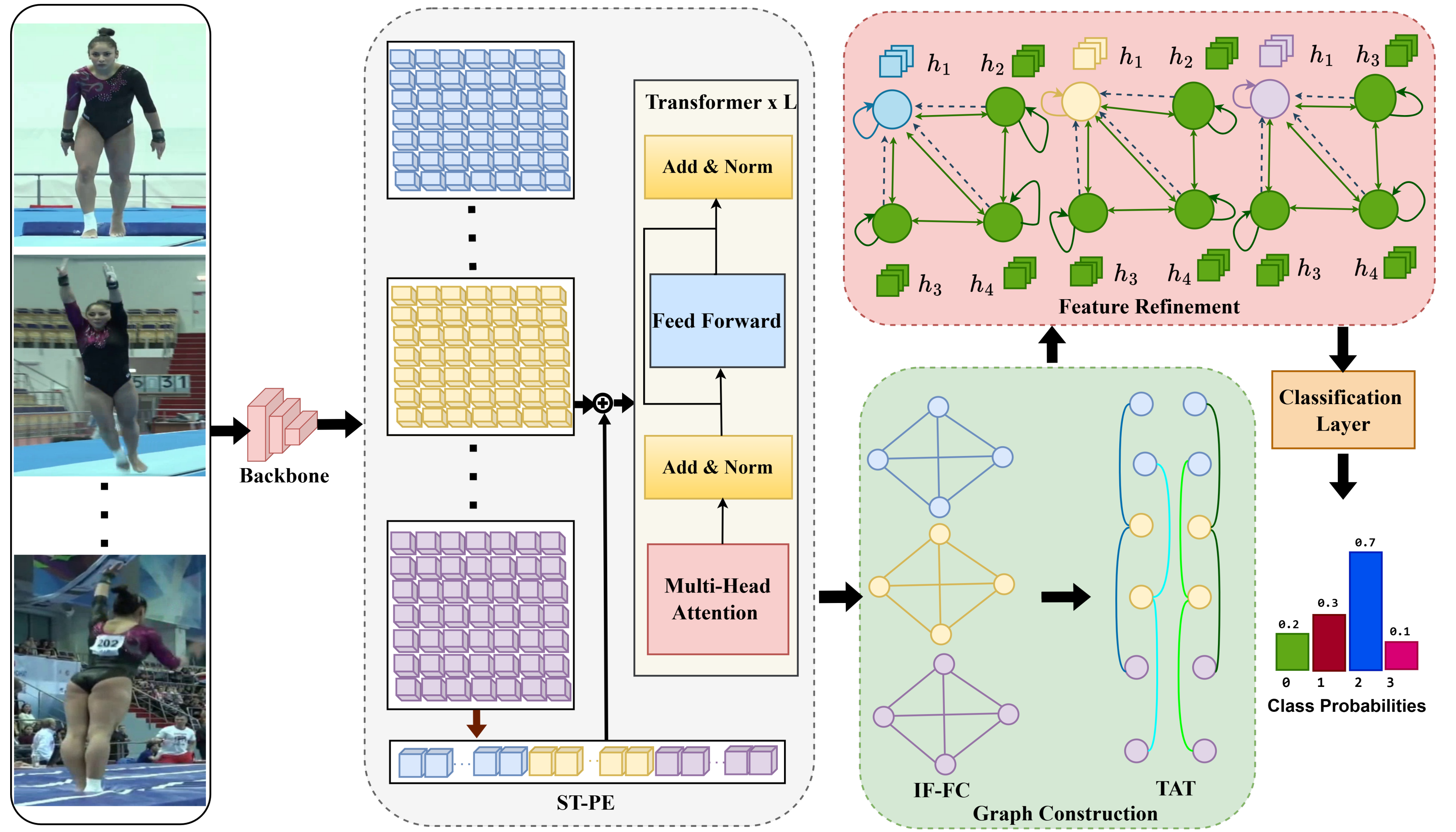
- Hybrid Transformer–Graph** for global + local reasoning
- RGB-only SotA** — no pose, text, or flow
- Lightweight**: 15.7× fewer params than PEVL
- Plug-and-play**: I3D, SlowFast, CSN, R(2+1)D

5. Efficiency & Complexity

Model	Params	GFLOPs
PGVT	265M	222
PEVL	1095M	510
TAG-Head	69.9M	153.4

15.7× smaller than PEVL

2. Proposed Approach



- Backbone + ST-PE**: 3D tokens $\mathbf{z}_{t,p} = \mathbf{x}_{t,p} + \mathbf{e}_{t,p}$ with learnable space-time positional encodings.
- Transformer Encoder**: multi-head attention captures long-range spatio-temporal dependencies (Eq. a).
- Time-Aligned Graph**: **IF-FC** (intra-frame) sharpens spatial contrasts; **TAT** (temporal) stabilises motion; APPNP propagates without extra parameters (Eq. b).
- Classifier**: global spatio-temporal pooling followed by a linear head.

$$\text{MHA}(\mathbf{Z}) = \text{Concat}_{h=1}^H \left[\text{softmax} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} \right) \mathbf{V}_h \right] \mathbf{W}_o \quad (\text{a})$$

$$\mathbf{H}^{(k+1)} = (1 - \alpha) \mathcal{A} \mathbf{H}^{(k)} + \alpha \mathbf{H}^{(0)} \quad (\text{b})$$

4. State-of-the-Art Performance

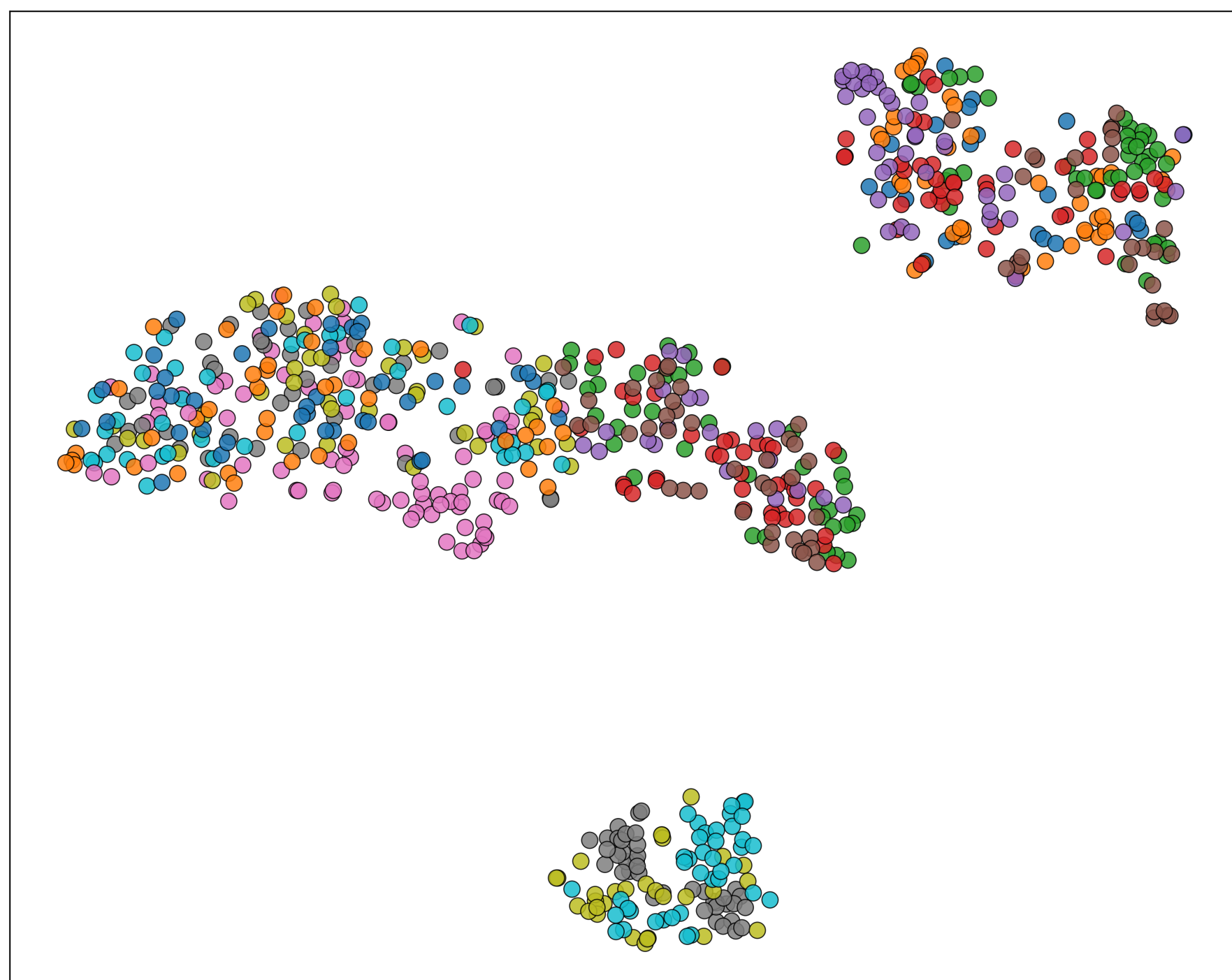
Method	Input	GYM99		GYM288		HAA500
		Top-1	MCA	Top-1	MCA	Top-1
SlowFast	R	93.9	90.6	86.8	51.2	25.1
TQN	R	93.8	90.6	89.6	61.9	—
VTCE	R+T	90.1	91.4	90.1	62.6	—
PGVT	R+P	96.7	91.6	91.0	63.6	—
PEVL	R+T+P	97.0	91.8	91.8	64.0	84.7
P2S+EVL	R+T+P	—	—	—	—	80.9
TAG-Head	R	95.6	93.8	92.2	68.6	86.1

R: RGB T: Text P: Pose

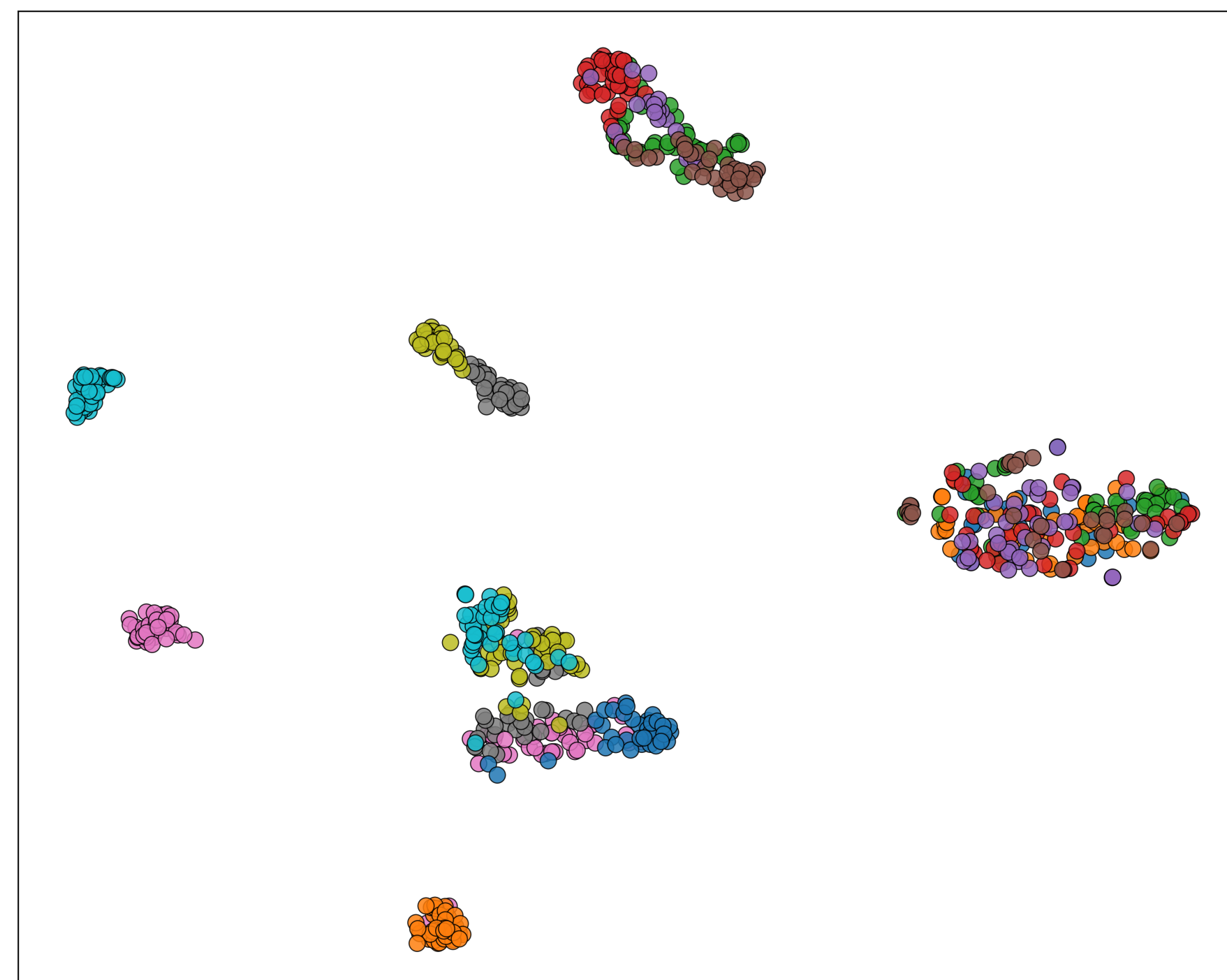
MCA gain on GYM288
+6.7%
over best RGB-only

HAA500 Top-1
+5.2%
over multimodal P2S+EVL

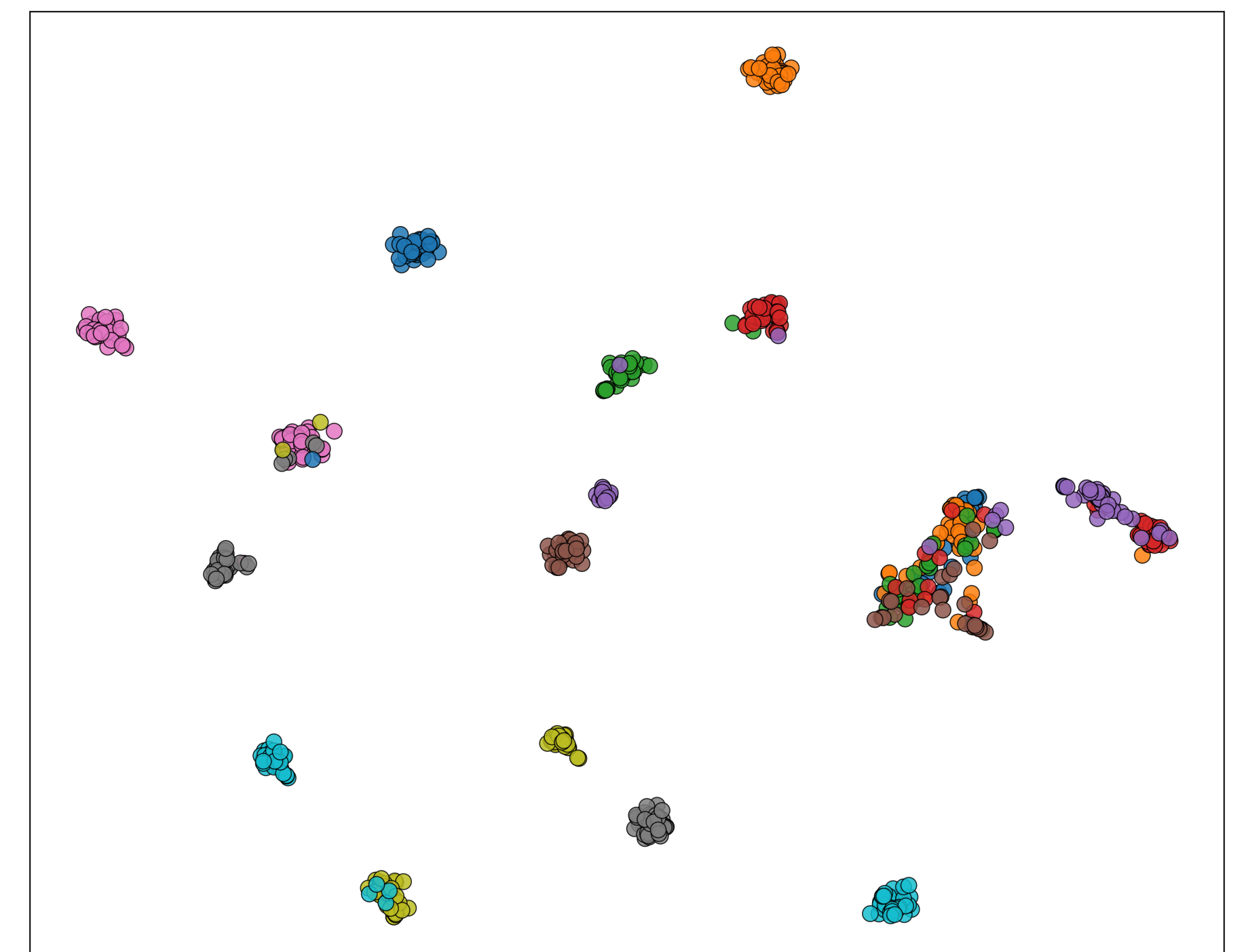
6. Feature Embedding Visualisation (GYM99)



(a) Transfer Learning



(b) Fine-tuning



(c) TAG-Head

t-SNE of learned features. (a) Features lack clear separation. (b) Clusters emerge but many classes remain entangled. (c) TAG-Head yields the most distinct, cohesive clusters.

7. Acknowledgements

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Code: github.com/ImtiazUlHassan/Tag-Head

8. References

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- Gasteiger, J., Bojchevski, A., Günnemann, S. Predict then propagate: GNNs meet personalized PageRank. *ICLR*, 2019.
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